Feature	Test Case	Expected Result (Pass)
CNIC Integrity	Enter 12 or 14 digits.	Field turns **Red**; Regex `^\d{13}$` blocks the "Sign Up" button.
Password Strength	Enter "12345" or "password".	Error: "Must contain 1 uppercase, 1 number, and 1 symbol."
File Size Limit	Upload a 2MB high-res photo.	Client-side error: "Image too large (Max 50KB)."
XSS Protection	Enter `<script>alert('hack')</script>` in the Name field.	The system must sanitize the input and store it as plain text, not executable code.
Bcrypt Hashing	Check a user in MongoDB Atlas.	The password must be a long, unreadable hash, **never** plain text.

2. Integration Testing: The Role-Based Logic

These tests verify the "Branching" logic for Elders, Caregivers, and Volunteers.

CASE 1: ELDER (The Safety Logic)

- **Test Case (Family Link):** Select "Lives Alone: No."
  - **Pass Criteria:** 1-3 Family contact fields must render immediately. If 0 contacts are filled, the Signup button must stay disabled.
- **Test Case (Medical "Other"):** Select a disorder, then select "Other."
  - **Pass Criteria:** A text area must appear. If left empty, it must turn **Red** upon submission attempt.
- **Test Case (Location Modal):** Click the Elder signup button.
  - **Pass Criteria:** The custom **Explanation Modal** must appear **before** the browser's native location prompt.

CASE 2: CAREGIVER (The Handshake Logic)

- **Test Case (Pairing Security):** Enter a correct Senior Email but a random 6-digit code.
    - **Pass Criteria:** System returns "Invalid Pairing Code." Account creation is blocked to prevent unauthorized data access.
- **Test Case (Session Persistence):** Log in as a Caregiver, then refresh the page.
    - **Pass Criteria:** The JWT (Json Web Token) must keep the user logged in without asking for credentials again.

CASE 3: VOLUNTEER (The Proximity Logic)

- **Test Case (Location Block):** Click "Deny" on the Location Modal.
    - **Pass Criteria:** Signup is blocked. Warning: "Location access is mandatory for volunteer duties."
- **Test Case (Radius Verification):** Set Radius to 5km.
    - **Pass Criteria:** Verify the database stores the `service_radius` and a `2dsphere` location point correctly.

3. Strategic Suggestions for "Unrejectable" Security

To make this project stand out to your SZABIST supervisors, instruct Openclaw to implement these:

1. **Rate Limiting:** If someone tries to guess the "6-digit Pairing Code" more than 5 times, block their IP for 15 minutes. This prevents "Brute Force" attacks on senior data.
2. **Environment Variables:** Ensure `MONGO_URI` and `JWT_SECRET` are **never** in the code. They must stay in the `.env` file. If Openclaw pushes a `.env` file to GitHub, it is a **Fail**.
3. **Data Minimization:** Only store what you need. For example, don't store the user's full date of birth if you only need their "Age" for the medical module.
4. **The "Red Indicator" UX:** Ensure that when a field turns red, a small "!" icon or text appears for color-blind users (Accessibility)

4. System Testing: The Final Handshake

1. **End-to-End Flow:** * Senior Signs up -> Generates Code.
    - Caregiver Signs up -> Enters Code.
    - **Verification:** Log in as Caregiver. Can you see the Senior's name? If yes, the **Integration** is successful.
2. **OTP Verification:** Test the "Forgot Password" flow. Ensure the email arrives and the OTP expires after 5 minutes for security.